# FIITJEE

# ALL INDIA TEST SERIES

## FULL TEST – XVII

## JEE (Main)-2021

## TEST DATE: 14-08-2021

**Time Allotted: 3 Hours** **Maximum Marks: 300**

**General Instructions:**

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts.**
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A and Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B :** Do any five questions out of 10 Questions.

  **Section-A (01 – 20, 31 – 50, 61 – 80)** *contains 60 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

  ***Section-B (21 – 30, 51 – 60, 81 – 90)*** *contains 30 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+4 marks*** *for correct answer. There is no negative marking.*

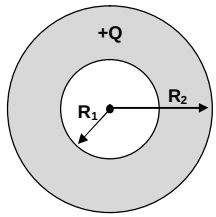
# Physics

## PART – A

### SECTION – A
### (One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

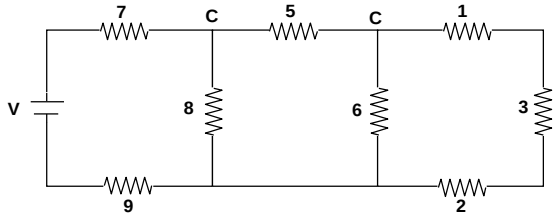
1. Figure shows a thick metallic sphere. If it is given a charge +q, then electric field will be present in the region.




(A) $r < R_1$ only
(B) $r > R_2$ only
(C) $r < R_1$ and $r > R_2$
(D) $R_1 < r < R_2$

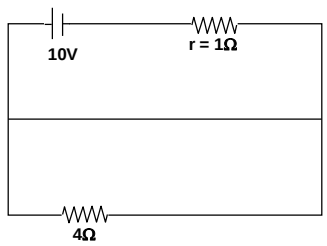
2. A capacitor of capacitance 10 $\mu$F is charged to a potential 50 V with a battery. The battery is now disconnected and an additional charge 200 $\mu$C is given to the positive plate of the capacitor. The potential difference across the capacitor will be.
(A) 50 V
(B) 80 V
(C) 100 V
(D) 60 V

3. Each resistance shown in the network is in ohm. Current through the resistor 3 $\Omega$ is 0.25 A. The input voltage 'V' is equal to




(A) 10 V
(B) 20 V
(C) 5 V
(D) $\frac{15}{2}$ V

4. Potential difference across the terminals of the battery shown in figure is




(A) 8 V
(B) 10 V
(C) 6 V
(D) zero
(r = internal resistance of battery)

5. A charged particle enters a uniform magnetic field with velocity vector at an angle of $45^o$ with the magnetic field. The pitch of the helical path followed by the particle is p. the radius of the helix will be.

(A) $\frac{p}{\sqrt{2}\pi}$

(B) $\sqrt{2}p$

(C) $\frac{p}{2\pi}$

(D) $\frac{\sqrt{2}p}{\pi}$

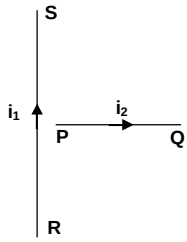
6. A current carrying wire PQ is placed near an another long current carrying wire RS. If free to move, wire PQ will have

(A) translational motion only

(B) rotational motion only

(C) translational as well as rotational motion

(D) neither translational nor rotational motion



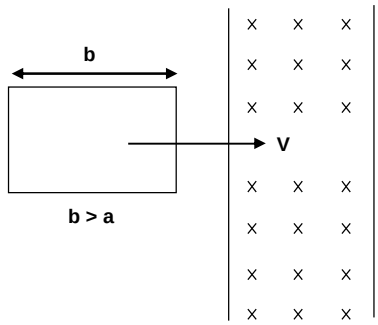

7. In the given arrangement, the loop is moved with constant velocity v in a uniform magnetic field B in a restricted region of width a. The time for which the emf is induced in the circuit is

(A) $\frac{2b}{v}$

(B) $\frac{2a}{v}$

(C) $\frac{(a+b)}{v}$

(D) $\frac{2(b-a)}{v}$



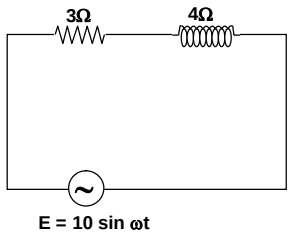

8. An ac-circuit having supply voltage E consists of a resistor of resistance $3\Omega$ and an inductor of reactance $4\Omega$ as shown in the figure. The voltage across the resistance at $t = \pi/\omega$ is

(A) 6.4 V

(B) 10 V

(C) zero

(D) 4.8 V



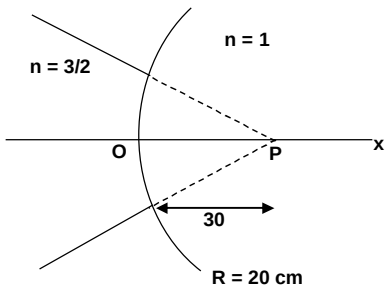

9. The image for the converging beam after refraction through the curved surface is formed at

(A) 40 cm

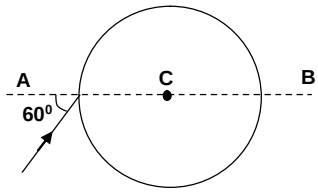
(B) $\frac{40}{3}$cm

(C) 20 cm

(D) $\frac{180}{7}$cm



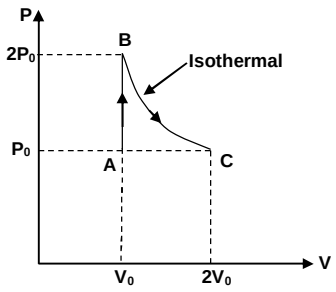
10. A ray of light falls on a transparent sphere with centre at C as shown in figure. The ray emerges from the sphere parallel to line AB. The refractive index of the sphere is.




(A) $\sqrt{2}$
(B) $\sqrt{3}$
(C) 3/2
(D) 4/3

11. If a minima is Young's double slit experiment occurs directly infront of one of the slits (distance between slit and screen D = 12 cm and distance between slits d = 5 cm) then the wavelength of the radiation used is
(A) 2 cm only
(B) 4 cm only
(C) $2\text{m}, \frac{2}{3}\text{cm}, \frac{2}{5}\text{cm}$
(D) $4\text{cm}, \frac{4}{3}\text{cm}, \frac{4}{5}\text{cm}$

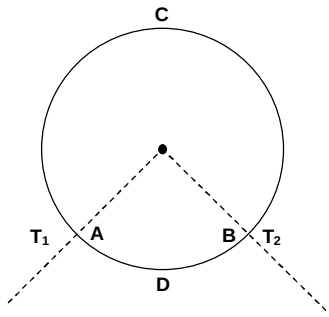
12. Diatomic ideal gas undergoes a thermodynamic change according to the P-V diagram shown in the figure. The total heat given to the gas is nearly (use in 2 = 0.7)




(A) $2.5P_0V_0$
(B) $1.4P_0V_0$
(C) $3.9P_0V_0$
(D) $1.1P_0V_0$

13. A ring consisting of two parts ADB and ACB of same conductivity K carries an amount of heat H. The ADB part is now replaced with another metal keeping the temperatures $T_1$ and $T_2$ constant. The heat carried increases to 2H. What should be the conductivity of the new ADB part?

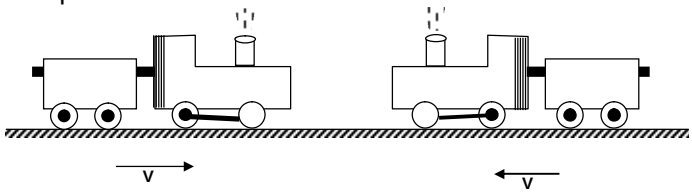
$\left(\text{Given}\frac{\text{ACB}}{\text{ADB}} = 3\right)$




(A) $\frac{7}{3}\text{K}$
(B) 2K
(C) $\frac{5}{2}\text{K}$
(D) 3K

14. A wire of density 9 gm/cm$^3$ is stretched between two clamps 1.00 m apart while subjected to an extension of 0.05 cm. The lowest frequency of transverse vibrations in the wire is (Assume Young's modulus $Y = 9\times10^{10}\,\text{N/m}^2$)
(A) 35 Hz
(B) 45 Hz
(C) 75 Hz
(D) 90 Hz

15. Two trains move towards each other with the same speed. Speed of sound is 340 $ms^{-1}$. If the frequency of the tone of the whistle of one when heard by the other changes by 9/8 times, then the speed of each train is.



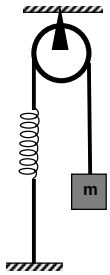

(A) 10 $ms^{-1}$

(B) 40 $ms^{-1}$

(C) 20 $ms^{-1}$

(D) 15 $ms^{-1}$

16. A mass m = 8 kg is attached to a spring as shown in figure and held in position so that the spring remains unstretched. The spring constant is 200 N/m. The mass m is then released and begins to undergo small oscillations. The maximum velocity of the mass will be (g = 10 $m/s^2$)



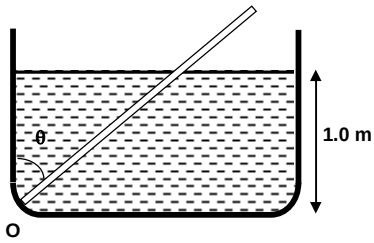

(A) 1 m/s

(B) 2 m/s

(C) 4 m/s

(D) 5 m/s

17. A uniform rod of length 2.0 m specific gravity 0.5 and mass 2 kg is hinged at one end to the bottom of a tank of water (specific gravity = 1.0) filled upto a height of 1.0 m as shown in figure. Taking the case $\theta \neq 0^0$ the force exerted by the hinge on the rod is (g = 10 $m/s^2$)




(A) 10.2 N upwards

(B) 4.2 N downwards

(C) 8.3 N downwards

(D) 6.2 N upwards

18. A uniform rod of length L has a mass per unit length $\lambda$ and area of cross-section A. The elongation in the rod is $\ell$ due to its own weight if it is suspended from the ceiling of a room. The Young's modulus of the rod is

(A) $\frac{2\lambda gL^2}{A\ell}$

(B) $\frac{\lambda gL^2}{2A\ell}$

(C) $\frac{2\lambda gL}{A\ell}$

(D) $\frac{\lambda g\ell^2}{AL}$

19. The potential at the surface of a planet of mass M and radius R is assumed to be zero. Choose the most appropriate option.

(A) The potential at infinity is $\frac{GM}{R}$

(B) The potential at the centre of planet is $-\frac{GM}{2R}$

(C) Both (A) and (B) are correct

(D) Both (A) and (B) are wrong

20. Photons with energy 5 eV are incident on a cathode C, on a photelectric cell. The maximum energy of the emitted photoelectrons is 2 eV. When photons of energy 6 eV are incident on C, no photoelectrons will reach the anode A if the stopping potential of A relative to C is

(A) 3 V

(B) –3 V

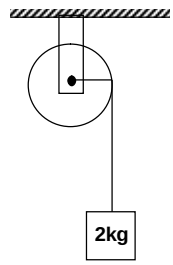
(C) –1 V

(D) 4 V

## SECTION – B
### (Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

21. A small pulley of radius 20 cm and moment of inertial 0.32 kg -$m^2$ is used to hang a 2 kg mass with the help of massless string. If the block is released, for no slipping condition find the acceleration of the block (in $m/s^2$).




22. Radius of gyration of a body about an axis at a distance 6 cm from its COM is 10 cm. Its radius of gyration about a parallel axis passing through its COM is (n) cm. Find value of n.

23. At high altitude, a body explodes at rest into two equal fragments with one fragment receiving horizontal velocity of 10 m/s. Find the time taken (in second) by the velocity vectors of the fragments to make $90^0$ ($g = 10\ m/s^2$).

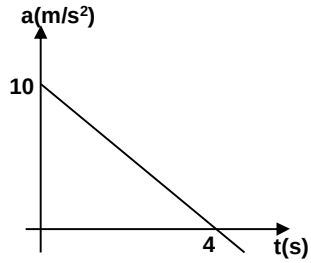
24. A small coin of mass 5 g is placed at a distance 5 cm from centre on a flat horizontal turn table. The turn table is observed to make 3 revolution in $\sqrt{10}$ s. The frictional force on the coin is n N. Find value of 10n.

25. An open knife edge of mass 200 g is dropped from height 5 m on a cardboard. If the knife edge penetrates a distance 2 m into the cardboard. Find the average resistance offered by the cardboard to the knife edge (in N). ($g = 10 m/s^2$)

26. A 4 kg block A is placed on the top of 8 kg block B which rests on a smooth table. A just slips on B when a force of 12 N is applied on A. Then the maximum horizontal force F applied on B to make both A and B move together, is (x) N. Find value of x.

27. An object is projected with a velocity of 20 m/s making an angle of $45^0$ with horizontal. The equation for trajectory is $h = Ax - Bx^2$ where h is height, x is horizontal distance. A and B are constants. The ratio A: B is x. Find value of x. ($g = 10\ m/s^2$).

28. The acceleration -time graph of a particle moving along a straight line is as shown in figure. At what time (in second) the particle acquires its initial velocity?




29. Light of wavelength 0.6 μm from a sodium lamp falls on a photocell and causes the emission of photoelectrons for which the stopping potential is 0.5 V. With light of wavelength 0.4μm from a sodium lamp, stopping potential is 1.5 V. With this data, the value of h/e is ($n \times 10^{-13}$) Vs. Find value of n.

30. In a certain circuit, current changes with time according to $i = 2\sqrt{t}$ . Root mean square value of current between t = 2 to t = 4s will be (x) A. Find value of x.

# Chemistry

## PART – B

### SECTION – A
### (One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

31. The triad of nuclei that is isotonic is
(A) $^{14}_{6}C$, $^{15}_{7}N$, $^{17}_{9}F$
(B) $^{12}_{6}C$, $^{15}_{7}N$, $^{19}_{9}F$
(C) $^{14}_{6}C$, $^{14}_{7}N$, $^{17}_{9}F$
(D) $^{14}_{6}C$, $^{14}_{7}N$, $^{19}_{9}F$

32. The correct order of second ionization potential of carbon, nitrogen, oxygen and fluorine is
(A) C > N > O > F
(B) O > N > F > C
(C) O > F > N > C
(D) F > O > N > C

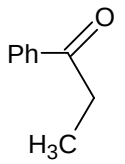
33. Which of the following is paramagnetic?
(A) $O_2^-$
(B) $CN^-$
(C) CO
(D) $NO^+$

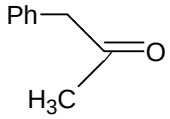
34. Which one among the following pairs of ions cannot be separated $H_2S$ in dilute hydrochloric acid?
(A) $Bi^{3+}, Sn^{4+}$
(B) $Al^{3+}, Hg^{2+}$
(C) $Zn^{2+}, Cu^{2+}$
(D) $Ni^{2+}, Cu^{2+}$

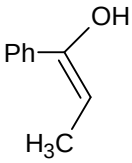
35. The volume strength of 1.5N $H_2O_2$ solution is
(A) 4.8
(B) 8.4
(C) 3.0
(D) 8.0

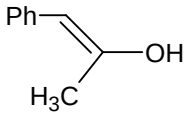
36. Arrange the following compounds in order of increasing dipole moment.
Toluene (I) m-dichlorobenzene (II)
o-dichlorobenzene (III) p-dichlorobenzene (IV)
(A) I < IV < II < III
(B) IV < I < II < III
(C) IV < I < III < II
(D) IV < II < I < III

37. $Ph-C\equiv C-CH_3 \xrightarrow{Hg^{2+}/H^+} A$. A is

(A) 

(B) 

(C) 

(D) 

38. Nitrogen dioxide cannot be obtained by heating
(A) $KNO_3$
(B) $Pb(NO_3)_2$
(C) $Cu(NO_3)_2$
(D) $AgNO_3$

39. Amongst $Ni(CO)_4$, $[Ni(CN)_4]^{2-}$ and $NiCl_4^{2-}$

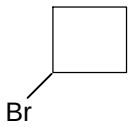
(A) $Ni(CO)_4$ and $NiCl_4^{2-}$ are diamagnetic and $[Ni(CN)_4]^{2-}$ is paramagnetic

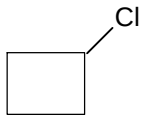
(B) $NiCl_4^{2-}$ and $[Ni(CN)_4]^{2-}$ are diamagnetic and $Ni(CO)_4$ is paramagnetic

(C) $Ni(CO)_4$ and $[Ni(CN)_4]^{2-}$ are diamagnetic and $NiCl_4^{2-}$ is paramagnetic

(D) $Ni(CO)_4$ is diamagnetic and $NiCl_4^{2-}$ and $[Ni(CN)_4]^{2-}$ are paramagnetic

40. What would be the product formed when 1–bromo–3–chlorocyclobutane reacts with two equivalents of metallic sodium in ether?

(A) 

(B) 

(C) 

(D) 

41. For the identification of $\beta$ – naphthol using dye test, it is necessary to use
(A) Dichloromethane solution of $\beta$ – naphthol
(B) Acidic solution of $\beta$ – naphthol
(C) Neutral solution of $\beta$ – naphthol
(D) Alkaline solution of $\beta$ – naphthol

42. In the Cannizzaro reaction, $2PhCHO \xrightarrow{^-OH} PhCH_2OH + PhCO_2^-$, the slowest step is
(A) the attack of $OH^-$ at the carbonyl group,
(B) the intermolecular transfer of hydride to the carbonyl group,
(C) the abstraction of proton from the carboxylic acid,
(D) the deprotonation of $PhCH_2OH$.

43. In the reaction p-chlorotoluene with $KNH_2$ in liq. $NH_3$ the major product is:
(A) o – toluidine
(B) m – toluidine
(C) p – toluidine
(D) p – chloroaniline

44. Among cellulose, polyvinyl chloride, nylon and natural rubber, the polymer in which the intermolecular force of attraction is weakest is
(A) Nylon
(B) Polyvinyl chloride
(C) Cellulose
(D) Natural Rubber

45. In van der Waals equation of state for a non-ideal gas, the term that accounts for intermolecular forces is
(A) $\left(V - b\right)$
(B) RT
(C) $\left(P + \frac{a}{V^2}\right)$
(D) $\left(RT\right)^{-1}$

46. The difference between heats of reaction at constant pressure and constant volume for the reaction: $2C_6H_6(\ell) + 15O_2(g) \rightarrow 12CO_2(g) + 6H_2O(\ell)$ at $25^oC$ in kJ is
(A) – 7.43
(B) +3.72
(C) – 3.72
(D) +7.43

47. The best indicator for detection of end point in titration of a weak acid and a strong base is:
(A) methyl orange (pH range: 3 to 4)
(B) methyl red (pH range: 5 to 6)
(C) bromothymol blue (pH range: 6 to 7.5)
(D) phenolphthalein (pH range: 8 to 9.6)

48. $^{27}_{13}Al$ is a stable isotope, $^{29}_{13}Al$ is expected to disintegrate by
(A) $\alpha$ – emission
(B) $\beta$ – emission
(C) positron emission
(D) proton emission

49. When mercuric iodide is added to the aqueous solution of potassium iodide then
(A) freezing point is raised.
(B) freezing point is lowered.
(C) freezing point does not change
(D) boiling point does not change.

50. The rusting of iron takes place as follows
$2H^+ + 2e^- + 1/2O_2(g) \rightarrow H_2O(\ell); E^o = +1.23V$
$Fe^{2+} + 2e^- \rightarrow Fe(s); E^o = -0.44V$
Calculate $\Delta G^o$ for the net process

(A) $-322$ kJ $mol^{-1}$
(B) $-161$ kJ $mol^{-1}$
(C) $-152$ kJ $mol^{-1}$
(D) $-76$ kJ $mol^{-1}$

## SECTION – B
### (Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

51. The dipole moment of KCl is $3.336 \times 10^{-29}$ Coulomb meters with indicates that it is a highly polar molecule. The interatomic distance between $K^+$ and $Cl^-$ in this molecule is $2.6 \times 10^{-10}$ m. Calculate the percentage ionic character of KCl. (Given that charge of electron = $1.602 \times 10^{-19}$ C)

52. 3.7 g of a gas at $25^oC$ occupied the same volume as 0.184g of hydrogen at $17^oC$ and at the same pressure. What is the molecular weight of the gas?

53. At room temperature, ammonia gas at 1 atm pressure and hydrogen chloride gas at P atm pressure are allowed to effuse through identical pin holes from opposite ends of a glass tube of one metre length and of uniform cross-section. Ammonium chloride is first formed at a distance of 60 cm from the end through which HCl gas is sent in. What is the value of P?

54. Estimate the average S–F bond energy (in kJ $mol^{-1}$) in $SF_6$. The values of standard enthalpy of formation of $SF_6$(g), S(g) and F(g) are: - 1100, 275 and 80 kJ $mol^{-1}$ respectively.

55. The pH of blood stream is maintained by a proper balance of $H_2CO_3$ and $NaHCO_3$ concentrations. What volume of 5M $NaHCO_3$ solution (in ml) should be mixed with a 10 ml sample of blood which is 2M in $H_2CO_3$ in order to maintain a pH of 7.4? $K_a$ for $H_2CO_3$ in blood is $7.8 \times 10^{-7}$. (Given that antilog 0.4 = 2.512)

56. A metal crystallises into two cubic phases, face centered cubic (FCC) and body centered cubic (BCC), whose unit cell lengths are 3.5 and 3.0 Å respectively. Calculate the ratio of densities of FCC and BCC.

57. A solution of a nonvolatile solute in water freezes at $-0.30^oC$. The vapour pressure of pure water at 298 K is 23.51 mm Hg and $K_f$ for water is 1.86 K kg $mol^{-1}$. Calculate the vapour pressure (in mm of Hg) of this solution at 298 K.

58. How long (in seconds) a current of 3 ampere has to be passed through a solution of silver nitrate to coat a metal surface of 80 $cm^2$ with a 0.005 mm thick layer? Density of silver is 10.5 $g/cm^3$. (Given that At. Wt. of Ag = 108 and F = 96500C $mol^{-1}$)

59. A first order reaction $A \rightarrow B$, requires activation energy of 70 kJ $mol^{-1}$. When a 20% solution of A was kept at $25^oC$ for 20 minutes, 25% decomposition took place. What will be the percent decomposition in the same time in a 30% solution maintained at $40^oC$? Assume that activation energy remains constant in this range of temperature. (Given that log 2 = 0.3, log 3 = 0.48, antilog 0.5879 = 3.872 & antilog 0.465 = 2.917)

60. Igniting $MnO_2$ converts it quantitatively to $Mn_3O_4$. A sample of pyrolusite is of the following composition: $MnO_2$ 80%, $SiO_2$ and other inter constituents 15%, rest being water. The sample is ignited in air to constant weight. What is the percentage of Mn in the ignited sample?
[At. Wt. of O = 16, Mn = 54.9]

# Mathematics

## PART – C

### SECTION – A
### (One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

61. Let S be the set of complex number z which satisfy $\log_{1/3}\left\{\log_{1/2}\left(|z|^2+4|z|+3\right)\right\}<0$, then S is (where $i=\sqrt{-1}$)
(A) $1\pm i$
(B) $3-i$
(C) $\frac{5}{2}+4i$
(D) empty set

62. For the equation $|x^2-2x-3|=b$ which statement or statements are true?
(A) for b < 0 there are no solutions
(B) for b = 0 there are three solution
(C) for 0 < b < 1 there are four solutions
(D) for b = 1 there are two solutions

63. If n is even positive integer, then the condition that the greatest term in the expansion of $(1+x)^n$ may have the greatest coefficient also is
(A) $\frac{n}{n+2}<x<\frac{n+2}{n}$
(B) $\frac{n+1}{n}<x<\frac{n}{n+1}$
(C) $\frac{n}{n+4}<x<\frac{n+4}{n}$
(D) None of these

64. If $f(x)=ax^2+bx+c, a,b,c\in R$ and equation $f(x)-x=0$ has imaginary roots $\alpha$ and $\beta$ and $\gamma,\delta$ the roots of $f\{f(x)\}-x=0$, then $\begin{vmatrix}2 & \alpha & \delta\\ \beta & 0 & \alpha\\ \gamma & \beta & 1\end{vmatrix}$ is
(A) 0
(B) purely real
(C) purely imaginary
(D) None of these

65. Domain of $\sin^{-1}[\sec x]$([.] is greatest integer less than or equal to x) is
(A) $\{(2n+1)\pi,(2n+9)\pi\}\cup\{[(2m-1)\pi,2m\pi+\pi/3),m\in I\}$
(B) $\{2n\pi,n\in I\}\cup\{[2m\pi,(2m+1)\pi,m\in I\}$
(C) $\{(2n+1)\pi,n\in I\}\cup\{[2m\pi,2m\pi+\pi/3),m\in I\}$
(D) None of the above

66. If x is a real number in [0, 1], then the value of $f(x) = \lim_{m\to\infty}\lim_{n\to\infty}\left\{1+\cos^{2m}\left(n!\pi x\right)\right\}$ is given by

(A) 2 or 1 according as x is rational or irrational
(B) 1 or 2 according as x is rational or irrational
(C) 1 or all x
(D) 2 or 1 for all x

67. If $f(x) = \lim_{n\to\infty}\frac{x^n - x^{-n}}{x^n + x^{-n}}, x > 1$, then $\int \frac{xf(x)\ln\left(x+\sqrt{\left(1+x^2\right)}\right)}{\sqrt{\left(1+x^2\right)}}dx$ is

(A) $\ln\left(x+\sqrt{\left(1+x^2\right)}\right) - x + c$

(B) $\frac{1}{2}\left\{x^2\ln\left(x+\sqrt{\left(1+x^2\right)}\right) - x^2\right\} + c$

(C) $x\ln\left(x+\sqrt{\left(1+x^2\right)}\right) - \ln\left(x+\sqrt{\left(1+x^2\right)}\right) + c$

(D) None of the above

68. The area bounded by $y = 2 - |2 - x|, y = \frac{3}{|x|}$ is

(A) $\left(\frac{5-4\ln 2}{3}\right)$ sq unit

(B) $\left(\frac{2-\ln 3}{2}\right)$ sq unit

(C) $\left(\frac{4-3\ln 3}{2}\right)$ sq unit

(D) None of these

69. Solution of the differential equation $\sin y\frac{dy}{dx} = \cos y\left(1 - x\cos y\right)$ is

(A) $\sec y = x - 1 - ce^x$

(B) $\sec y = x + 1 + ce^x$

(C) $\sec y = x + e^x + c$

(D) None of these
(where c is arbitrary constant)

70. The line x + y = a meets the axes of x and y at A and B respectively, A triangle AMN is inscribed in the triangle OAB, O being the origin, with right angle at N, M and N lie respectively on OB and AB. If the area of the triangle AMN is $\frac{3}{8}$ of the area of the triangle OAB, then $\frac{AN}{BN}$ is equal to

(A) 1
(B) 2
(C) 3
(D) 4

71. Two sides of a rhombus are along the lines, x – y + 1 = 0 and $7x-y-5=0$, If its diagonals intersect at $(-1,-2)$, then which one of the following is a vertex of this rhombus?

(A) $\left(\frac{1}{3},-\frac{8}{3}\right)$

(B) $\left(-\frac{10}{3},-\frac{7}{3}\right)$

(C) $(-3,-9)$

(D) $(-3,-8)$

72. A circle touches the x-axis and also touches the circle with centre at (0, 3) and radius 2. The locus of the centre of the circle is.

(A) an ellipse
(B) a circle
(C) a hyperbola
(D) a parabola

73. The shortest distance between the parabolas $2y^2=2x-1$ and $2x^2=2y-1$ is

(A) $\frac{1}{2\sqrt{2}}$

(B) $\frac{1}{2}$

(C) $2\sqrt{2}$

(D) 4

74. A tangent to the ellipse $\frac{x^2}{25}+\frac{y^2}{16}=1$ at any point P meet the line x = 0 at a point Q. Let R be the image of Q in the line y = x, then circle whose extremities of a diameter are Q and R passes through a fixed point. The fixed point is

(A) (3, 0)
(B) (4, 0)
(C) (5, 0)
(D) (0, 0)

75. The equation to the plane through the point $(2,-1,0),(3,-4,5)$ parallel to a line with direction cosines proportional to 2, 3, 4 is 9x – 2y – 3z = k, where k is

(A) 20
(B) – 20
(C) 10
(D) – 10

76. If $0<\alpha<\pi/6$ and $\sin\alpha+\cos\alpha=\sqrt{7}/2$, then $\tan\alpha/2$ is equal to

(A) $\frac{\sqrt{7}-2}{3}$

(B) $\frac{\sqrt{7}+2}{3}$

(C) $\frac{\sqrt{7}}{3}$

(D) None of these

77. The solution of the equation $\sin^{10} x + \cos^{10} x = \frac{29}{16}\cos^4 2x$ is

(A) $x = \frac{n\pi}{4} + \frac{\pi}{8}, n \in I$

(B) $x = n\pi + \frac{\pi}{4}, n \in I$

(C) $x = 2n\pi + \frac{\pi}{2}, n \in I$

(D) None of these

78. If in $\Delta ABC, \frac{a\cos A + b\cos B + c\cos C}{a\sin B + b\sin C + c\sin A} = \frac{a+b+c}{9R}$, then the triangle ABC is

(A) isosceles
(B) equilateral
(C) right angled
(D) None of these

79. Let $\vec{r}, \vec{a}, \vec{b}$ and $\vec{c}$ be four non zero vectors such that $\vec{r}.\vec{a} = 0$, $\left|\vec{r} \times \vec{b}\right| = |\vec{r}||\vec{b}|$, $|\vec{r} \times \vec{c}| = |\vec{r}||\vec{c}|$, then $\left[\vec{a}\ \vec{b}\ \vec{c}\right]$ is equal to

(A) – 1
(B) 0
(C) 1
(D) 2

80. The value of $\lambda$ so that the points P, Q, R, S on the sides OA, OB, OC and AB of a regular tetrahedron are coplanar. When $\frac{OP}{OA} = \frac{1}{3}; \frac{OQ}{OB} = \frac{1}{2}, \frac{OR}{OC} = \frac{1}{3}$ and $\frac{OS}{AB} = \lambda$ is equal to

(A) $\lambda = \frac{1}{2}$
(B) $\lambda = -1$
(C) $\lambda = 0$
(D) for no value of $\lambda$

## SECTION – B
## (Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

81. If $y = 2[x] + 3 = 3[x-2] + 5$, then $[x+y]$ is ([x] denotes the integral part of x)

82. The coefficient of $x^{203}$ in the expansion of $(x-1)(x^2-2)(x^3-3).....(x^{20}-20)$ is

83. The number of non-negative integral solutions of $x_1 + x_2 + x_3 + 4x_4 = 20$ is

84. Let $a_r = r\left({}^7C_r\right), b_r = (7-r)\left({}^7C_r\right)$ and $A_r \begin{pmatrix} a_r & 0 \\ 0 & b_r \end{pmatrix}$. If $A = \sum_{r=0}^{7} A_r = \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$, then the value of a + b must be

85. An artillery target may be either at point 1 with probability $\frac{8}{9}$ or at point II with probability $\frac{1}{9}$. We have 21 shells each of which can be fired either at point I or II. Each shell may hit the target independently of the other shell with probability $\frac{1}{2}$. If the number of shells must be fired at point I to hit the target with maximum probability is x, then the value of 12x, must be

86. The least period of the function $\sin\left(\frac{\pi[x]}{12}\right)+\cos\left(\frac{\pi x}{4}\right)+\tan\left(\frac{\pi[x]}{3}\right)$ is $\lambda$, then the value of $201\lambda$ must be (when [.] denotes the greatest integer function)

87. If $f(x)+f(y)=f\left(\frac{x+y}{1-xy}\right)$ for all $x,y\in R$ and $xy\neq 1$ and $\lim\limits_{x\to 0}\frac{f(x)}{x}=2$, then the value of $\frac{1500}{\pi}\frac{f\left(\sqrt{3}\right)}{f'(-2)}$ must be

88. If $\phi(x)$ is larger of $\left(e^x-1\right)$ and $(1+x)\ln(1+x)$ for $x\in(0,\infty)$, then the value of $502\ln\left(\phi(4)+1\right)$ must be

89. A wire of given length $\ell$ is cut into two portions which are bent into the shapes of a circle and a square respectively. If the sum of the areas of the circle and the square will be least when 9side of square) = $\lambda$ (diameter of circle), then the value of $1005\lambda$ must be

90. Let $f:R^+\to R$ be a differentiable function with $f(1)=3$ and satisfying $\int_1^{xy}f(t)dt=y\int_1^x f(t)+x\int_1^y f(t)dt\ \forall\ x,y\in R^+$, then the value of $f\left(e^{37}\right)$ must be